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Chapter 10 - Higher-Order Constant-Coefficient Equations

Chapter 9 developed the characteristic-equation method for second-order homogeneous equations. This chapter extends the same idea to any finite order.

We study equations of the form:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0, \quad a_n \neq 0,$$

where all coefficients are real constants.

The central questions are:

- how does the exponential trial function work at order n ?
- why must an n th-order solution contain n independent directions?
- what does root multiplicity contribute to the basis?
- how are repeated complex roots converted into real functions?
- how can a complicated root list be organised without losing a solution?
- how do n initial conditions determine n constants?
- when do known solutions reveal only part of an equation?

The chapter's organising idea is:

treat every characteristic root as an instruction for building basis functions, then count those functions until the differential equation's order has been matched.

All examples, data, and exercises are independently constructed. The reference text is used only to identify the mathematical topics.

Extending The Characteristic Equation To Order n

The Goal Of This Block

The goal is to derive the characteristic polynomial for an n th-order constant-coefficient homogeneous equation and make the notation readable.

Reading The Higher-Order Notation

The general equation is:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0.$$

Here:

- x is the independent variable
- $y = y(x)$ is the unknown function
- $y^{(k)}$ is the k th derivative, not the k th power of y
- $a_0, a_1, \dots, a_n$ are real constants
- $a_n \neq 0$ ensures that the equation really has order n

For example:

$$y^{(5)} - 4y''' + 7y' - 2y = 0$$

is fifth-order because its highest derivative is $y^{(5)}$.

The missing $y^{(4)}$ and y'' terms have coefficient zero:

$$y^{(5)} + 0y^{(4)} - 4y''' + 0y'' + 7y' - 2y = 0.$$

Making zero coefficients visible prevents terms from being assigned to the wrong power in the characteristic polynomial.

The Exponential Trial At Any Order

Use:

$$y = e^{rx},$$

where r is a constant to be determined.

Successive derivatives preserve the exponential factor:

$$\begin{aligned} y' &= r e^{rx}, \\ y'' &= r^2 e^{rx}, \\ y''' &= r^3 e^{rx}, \\ &\vdots \\ y^{(n)} &= r^n e^{rx}. \end{aligned}$$

The general pattern is:

$$\boxed{y^{(k)} = r^k e^{rx}}.$$

Every derivative contains the same nonzero factor e^{rx} . Only the power of r changes.

Deriving The Degree- n Polynomial

Start from:

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0.$$

For $y = e^{rx}$, use $y^{(k)} = r^k e^{rx}$. Substitution gives:

$$a_n r^n e^{rx} + a_{n-1} r^{n-1} e^{rx} + \cdots + a_1 r e^{rx} + a_0 e^{rx} = 0.$$

Factor out the common exponential:

$$e^{rx} (a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0) = 0.$$

Since $e^{rx} \neq 0$, the polynomial factor must be zero:

$$\boxed{a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = 0}.$$

This is the **characteristic equation**. Its left side is the **characteristic polynomial**.

A Fifth-Order Translation Example

Consider:

$$2y^{(5)} - 3y^{(4)} + 7y'' - 11y = 0.$$

Write all derivative orders, including the missing ones:

$$2y^{(5)} - 3y^{(4)} + 0y''' + 7y'' + 0y' - 11y = 0.$$

Replace each derivative order by the matching power of r :

$$\boxed{2r^5 - 3r^4 + 0r^3 + 7r^2 + 0r - 11 = 0}.$$

Omitting the zero terms gives:

$$2r^5 - 3r^4 + 7r^2 - 11 = 0.$$

The order of the ODE matches the degree of the characteristic polynomial.

The Scope Check Still Matters

The method applies directly only when the equation is linear, homogeneous, and constant-coefficient.

For example:

$$(1 + x)y''' + 2y = 0$$

has a variable coefficient.

The equation:

$$y^{(4)} + y^2 = 0$$

is nonlinear because of y^2 .

The equation:

$$y^{(4)} - 3y'' = e^x$$

is nonhomogeneous. This chapter can solve its associated homogeneous equation, but a particular solution is still needed for the complete forced problem.

Block 1 Summary

For:

$$a_n y^{(n)} + \cdots + a_1 y' + a_0 y = 0,$$

For the equation $a_n y^{(n)} + \cdots + a_1 y' + a_0 y = 0$, the trial $y = e^{rx}$ produces:

$$a_n r^n + \cdots + a_1 r + a_0 = 0.$$

Derivative order becomes polynomial power. Zero coefficients should be tracked even when their terms are not written.

Roots, Multiplicity, And The Basis Count

The Goal Of This Block

The goal is to connect polynomial degree to the number of functions required in the general solution.

an n th-degree characteristic polynomial supplies n roots when multiplicity is counted, and those roots must produce n independent real basis functions.

Roots Counted With Multiplicity

A degree- n polynomial has n complex roots when repeated roots are counted according to multiplicity.

For example:

$$p(r) = (r - 2)^3(r + 5)(r^2 + 4).$$

The factors give:

$$\begin{aligned} r &= 2 && \text{with multiplicity } 3, \\ r &= -5 && \text{with multiplicity } 1, \\ r &= 2i && \text{with multiplicity } 1, \\ r &= -2i && \text{with multiplicity } 1. \end{aligned}$$

Count the multiplicities:

$$3 + 1 + 1 + 1 = 6.$$

The polynomial is sixth-degree, so the count agrees.

The Root Ledger

A **root ledger** records what each root contributes.

For $p(r) = (r - 2)^3(r + 5)(r^2 + 4)$:

Root or pair	Multiplicity	Number of real modes
2	3	3
-5	1	1
$\pm 2i$	1	2

The total is:

$$3 + 1 + 2 = 6.$$

The practical meaning is:

do not stop when the roots have been found; translate every root and count every resulting basis function.

Distinct Real Roots

If the roots $r_1, \dots, r_n$ are real and distinct, each contributes one exponential.

The general solution is:

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x} + \dots + c_n e^{r_n x}.$$

There are n arbitrary constants because the n exponentials are linearly independent.

Repeated Real Roots

If $r = \rho$ has multiplicity s , it contributes:

$$e^{\rho x}, x e^{\rho x}, x^2 e^{\rho x}, \dots, x^{s-1} e^{\rho x}.$$

For example, $r = -3$ with multiplicity 4 contributes:

$$e^{-3x}, x e^{-3x}, x^2 e^{-3x}, x^3 e^{-3x}.$$

The powers begin at 0 and stop at one less than the multiplicity.

Complex-Conjugate Pairs

For real coefficients, $\alpha + i\beta$ is accompanied by $\alpha - i\beta$.

A simple pair contributes:

$$e^{\alpha x} \cos(\beta x), \quad e^{\alpha x} \sin(\beta x).$$

If the pair has multiplicity s , multiply both modes by:

$$1, x, x^2, \dots, x^{s-1}.$$

Thus a conjugate pair of multiplicity s contributes $2s$ real functions.

A Complete Formula Map

Each root instruction can be summarised as:

Root information	Real basis contribution
$r = \rho$, multiplicity s	$x^j e^{\rho x}, \quad j = 0, \dots, s-1$
$r = \alpha \pm i\beta$, multiplicity s	$x^j e^{\alpha x} \cos(\beta x), \quad j = 0, \dots, s-1$ $x^j e^{\alpha x} \sin(\beta x), \quad j = 0, \dots, s-1$

After using the map, count the basis functions. Their number must equal the order of the differential equation.

Block 2 Summary

Root multiplicity determines how many powers of x accompany a mode.

The completion check is:

$$\boxed{\text{number of independent real basis functions} = n}.$$

Factoring Higher-Degree Characteristic Polynomials

The Goal Of This Block

The goal is to factor common polynomial patterns without hiding the algebra. The root-to-solution rules become useful only after the roots are found.

Start With The Easiest Structure

Before trying a general formula, check for:

- a common factor r
- grouping
- a polynomial in r^2
- an obvious integer root
- a known factor supplied by the problem
- a perfect power such as $(r - \rho)^s$

High-degree polynomials do not always factor neatly by hand. When exact factorisation is unavailable, numerical roots may be needed. The root-to-basis rules remain the same.

Worked Cubic With Three Distinct Roots

Solve:

$$y''' - 3y'' - 6y' + 8y = 0.$$

Step 1: Write the characteristic equation. The derivative coefficients give:

$$r^3 - 3r^2 - 6r + 8 = 0.$$

Step 2: Find one root. Test $r = 1$:

$$\begin{aligned} p(1) &= 1^3 - 3(1)^2 - 6(1) + 8 \\ &= 1 - 3 - 6 + 8 \\ &= 0. \end{aligned}$$

Therefore $r = 1$ is a root and $(r - 1)$ is a factor.

Step 3: Expose the remaining quadratic. The cubic factors as:

$$r^3 - 3r^2 - 6r + 8 = (r - 1)(r^2 - 2r - 8).$$

Check by expansion:

$$\begin{aligned} (r - 1)(r^2 - 2r - 8) &= r^3 - 2r^2 - 8r - r^2 + 2r + 8 \\ &= r^3 - 3r^2 - 6r + 8. \end{aligned}$$

Step 4: Factor the quadratic. The remaining quadratic factors as:

$$r^2 - 2r - 8 = (r - 4)(r + 2).$$

Combining the factors $(r - 1)$, $(r - 4)$, and $(r + 2)$ gives:

$$(r - 1)(r - 4)(r + 2) = 0.$$

Step 5: List and classify the roots. Set each factor equal to zero:

$$r_1 = 1, \quad r_2 = 4, \quad r_3 = -2.$$

All three roots are real and distinct.

Step 6: Build and count the solution. Each root contributes one exponential:

$$y = c_1 e^x + c_2 e^{4x} + c_3 e^{-2x}.$$

The equation is third-order and the family contains three independent modes.

Worked Fourth-Order Equation In r^2

Solve:

$$y^{(4)} - 13y'' + 36y = 0.$$

Step 1: Write the characteristic equation. The missing y''' and y' terms have coefficient zero:

$$r^4 - 13r^2 + 36 = 0.$$

Step 2: Use a temporary variable. Only even powers appear. Let:

$$z = r^2.$$

With $z = r^2$, we also have $r^4 = z^2$, so:

$$z^2 - 13z + 36 = 0.$$

Step 3: Factor the quadratic in z . Find two numbers that multiply to 36 and add to -13 :

$$z^2 - 13z + 36 = (z - 4)(z - 9).$$

Therefore:

$$(z - 4)(z - 9) = 0.$$

Step 4: Return from z to r . Since $z = r^2$:

$$r^2 = 4 \quad \text{or} \quad r^2 = 9.$$

Solve both equations:

$$r = \pm 2 \quad \text{or} \quad r = \pm 3.$$

Step 5: Build and count the solution. The four roots are distinct and real. Therefore:

$$y = c_1 e^{2x} + c_2 e^{-2x} + c_3 e^{3x} + c_4 e^{-3x}.$$

Four roots produce four modes for the fourth-order equation.

A Common Factor r

Consider:

$$y^{(5)} + 4y''' = 0.$$

The characteristic equation is:

$$r^5 + 4r^3 = 0.$$

Factor out r^3 :

$$r^3(r^2 + 4) = 0.$$

The factor r^3 gives $r = 0$ with multiplicity 3. The factor $r^2 + 4$ gives $r = \pm 2i$.

The root ledger is:

Root or pair	Multiplicity	Real modes
0	3	$1, x, x^2$
$\pm 2i$	1	$\cos(2x), \sin(2x)$

Therefore:

$$y = c_1 + c_2x + c_3x^2 + c_4 \cos(2x) + c_5 \sin(2x).$$

The ledger contains $3 + 2 = 5$ real modes, matching the fifth-order equation.

Block 3 Summary

Factor before translating roots into functions. Useful structures include a common power of r , a polynomial in r^2 , and a known linear factor.

After factorisation:

1. list every root
2. record every multiplicity
3. convert roots to real basis functions
4. count the basis functions against the ODE order

Why Multiplicity Produces Powers Of x

The Goal Of This Block

The goal is to derive the repeated-root rule for any multiplicity.

every repeated factor $(r - \rho)$ removes one derivative from a multiplier; repeating the factor s times leaves a polynomial multiplier of degree at most $s - 1$.

Introduce The Differential Operator D

Let:

$$D = \frac{d}{dx}.$$

Then:

$$Dy = y', \quad D^2y = y'', \quad D^k y = y^{(k)}.$$

A characteristic factor $(r - \rho)$ corresponds to the differential operator:

$$(D - \rho).$$

For example, $(r - \rho)^3$ corresponds to $(D - \rho)^3$.

The Key Cancellation Identity

Let:

$$y = e^{\rho x} v(x),$$

where $v(x)$ is an unknown multiplier.

Differentiate using the product rule:

$$\begin{aligned} Dy &= \frac{d}{dx} (e^{\rho x} v) \\ &= \rho e^{\rho x} v + e^{\rho x} v'. \end{aligned}$$

Now apply $(D - \rho)$:

$$\begin{aligned} (D - \rho)y &= Dy - \rho y \\ &= (\rho e^{\rho x} v + e^{\rho x} v') - \rho e^{\rho x} v. \end{aligned}$$

The two terms containing $\rho e^{\rho x} v$ cancel:

$$\boxed{(D - \rho) (e^{\rho x} v) = e^{\rho x} v'}.$$

Applying $(D - \rho)$ removes the exponential's own derivative contribution and differentiates only the multiplier v .

Repeat The Operator s Times

Apply the identity once:

$$(D - \rho) (e^{\rho x} v) = e^{\rho x} v'.$$

Apply $(D - \rho)$ a second time:

$$(D - \rho)^2 (e^{\rho x} v) = e^{\rho x} v''.$$

Continuing the pattern gives:

$$\boxed{(D - \rho)^s (e^{\rho x} v) = e^{\rho x} v^{(s)}}.$$

If $(r - \rho)^s$ is the entire characteristic polynomial for this part of the equation, then:

$$(D - \rho)^s y = 0.$$

Substitute $y = e^{\rho x} v$:

$$e^{\rho x} v^{(s)} = 0.$$

Since the exponential is nonzero:

$$v^{(s)} = 0.$$

Solving $v^{(s)} = 0$

If the s th derivative of v is zero, then v is a polynomial of degree at most $s - 1$:

$$v(x) = c_1 + c_2x + c_3x^2 + \cdots + c_sx^{s-1}.$$

Substitute this into $y = e^{\rho x}v$:

$$y = e^{\rho x} (c_1 + c_2x + c_3x^2 + \cdots + c_sx^{s-1}).$$

Equivalently:

$$y = c_1e^{\rho x} + c_2xe^{\rho x} + \cdots + c_sx^{s-1}e^{\rho x}.$$

This proves the repeated-root rule.

Worked Fifth-Order Repeated-Root Example

Solve the equation whose characteristic polynomial is:

$$p(r) = (r - 2)^3(r + 3)^2.$$

Step 1: Record the roots and multiplicities. The factor $(r - 2)^3$ gives:

$$r = 2 \quad \text{with multiplicity 3.}$$

The factor $(r + 3)^2$ gives:

$$r = -3 \quad \text{with multiplicity 2.}$$

Step 2: Build the chain for $r = 2$. Multiplicity 3 requires powers x^0, x^1, x^2 :

$$e^{2x}, \quad xe^{2x}, \quad x^2e^{2x}.$$

Step 3: Build the chain for $r = -3$. Multiplicity 2 requires powers x^0, x^1 :

$$e^{-3x}, \quad xe^{-3x}.$$

Step 4: Combine and count. The general solution is:

$$y = e^{2x}(c_1 + c_2x + c_3x^2) + e^{-3x}(c_4 + c_5x).$$

The first root contributes 3 modes and the second contributes 2:

$$3 + 2 = 5.$$

The basis count matches the fifth-degree characteristic polynomial.

Expanded Differential Equation For The Example

To recover the differential equation, expand:

$$(r - 2)^3(r + 3)^2.$$

First calculate each power:

$$(r - 2)^3 = r^3 - 6r^2 + 12r - 8,$$

$$(r + 3)^2 = r^2 + 6r + 9.$$

Multiply the polynomials:

$$(r - 2)^3(r + 3)^2 = (r^3 - 6r^2 + 12r - 8)(r^2 + 6r + 9).$$

Distribute each term from the first polynomial:

$$\begin{aligned}(r - 2)^3(r + 3)^2 &= (r^5 + 6r^4 + 9r^3) \\ &\quad + (-6r^4 - 36r^3 - 54r^2) \\ &\quad + (12r^3 + 72r^2 + 108r) \\ &\quad + (-8r^2 - 48r - 72).\end{aligned}$$

Now combine equal powers of r :

$$\begin{aligned}(r - 2)^3(r + 3)^2 &= r^5 + (6 - 6)r^4 \\ &\quad + (9 - 36 + 12)r^3 \\ &\quad + (-54 + 72 - 8)r^2 \\ &\quad + (108 - 48)r - 72 \\ &= r^5 - 15r^3 + 10r^2 + 60r - 72.\end{aligned}$$

The absent r^4 coefficient is zero. Therefore the associated equation is:

$$\boxed{y^{(5)} - 15y''' + 10y'' + 60y' - 72y = 0}.$$

Block 4 Summary

The identity:

$$(D - \rho)^s(e^{\rho x}v) = e^{\rho x}v^{(s)}$$

Because $(D-\rho)^s(e^{\rho x}v) = e^{\rho x}v^{(s)} = 0$. Therefore v is a polynomial of degree at most $s-1$, which creates the chain:

$$e^{\rho x}, xe^{\rho x}, \dots, x^{s-1}e^{\rho x}.$$

Mixed Real And Complex Root Structures

The Goal Of This Block

The goal is to combine several root types in one solution without missing a conjugate partner or a multiplicity factor.

A Five-Step Translation Method

For a factored characteristic polynomial:

1. list every real root and its multiplicity
2. list every complex-conjugate pair and its multiplicity
3. write the real basis contribution from each ledger row
4. attach separate arbitrary constants
5. count the total number of real modes

Keep the root ledger visible until the basis count is complete.

Complete Mixed-Root Example

Suppose a fifth-order equation has characteristic polynomial:

$$p(r) = (r-2) \left((r+1)^2 + 4 \right)^2.$$

Step 1: Read the real root. The factor $(r - 2)$ gives:

$$r = 2 \quad \text{with multiplicity 1.}$$

It contributes:

$$e^{2x}.$$

Step 2: Read the complex pair. Set the quadratic factor equal to zero:

$$(r + 1)^2 + 4 = 0.$$

Subtract 4:

$$(r + 1)^2 = -4.$$

Take square roots:

$$r + 1 = \pm 2i.$$

Subtract 1:

$$r = -1 \pm 2i.$$

Because the quadratic factor is squared, the conjugate pair has multiplicity 2.

Step 3: Build the complex real basis. For $\alpha = -1$, $\beta = 2$, and multiplicity 2, use powers 1 and x on both trigonometric modes:

$$\begin{aligned} e^{-x} \cos(2x), \\ e^{-x} \sin(2x), \\ xe^{-x} \cos(2x), \\ xe^{-x} \sin(2x). \end{aligned}$$

Step 4: Combine the full family. Group the cosine and sine chains cleanly:

$$y = c_1 e^{2x} + e^{-x} [(c_2 + c_3 x) \cos(2x) + (c_4 + c_5 x) \sin(2x)].$$

Step 5: Count the modes. The real root contributes 1 mode. The repeated conjugate pair contributes $2 \times 2 = 4$ modes:

$$1 + 4 = 5.$$

The count matches the fifth-order equation.

Why The Sine Partner Cannot Be Omitted

Suppose a real-coefficient equation has the solution:

$$e^{3x} \cos(5x).$$

This mode comes from roots:

$$r = 3 \pm 5i.$$

The same conjugate pair necessarily supplies:

$$e^{3x} \sin(5x).$$

One trigonometric function alone does not represent the complete real basis for a conjugate pair.

Repeated Complex Pair Checklist

If $\alpha \pm i\beta$ has multiplicity s , check all four ingredients:

1. exponential envelope $e^{\alpha x}$
2. frequency β inside sine and cosine
3. both sine and cosine families
4. powers x^0 through x^{s-1} on both families

Missing any one ingredient produces an incomplete solution space.

Block 5 Summary

A simple real root contributes one mode. A real root of multiplicity s contributes s modes. A conjugate pair of multiplicity s contributes $2s$ real modes.

The sum of all ledger contributions must equal the equation's order.

Higher-Order Initial-Value Problems

The Goal Of This Block

The goal is to apply n initial conditions to an n th-order solution family without skipping derivative or substitution steps.

How Many Conditions Are Needed?

For a regular n th-order equation, an initial-value problem normally supplies:

$$y(x_0), y'(x_0), \dots, y^{(n-1)}(x_0).$$

These n values determine the n constants in the general homogeneous solution.

For a fourth-order equation, expect:

$$y(x_0), \quad y'(x_0), \quad y''(x_0), \quad y'''(x_0).$$

Initial data stop one derivative below the equation's order.

A Complete Fourth-Order Initial-Value Problem

Solve:

$$y^{(4)} + 4y'' = 0,$$

subject to:

$$y(0) = 6, \quad y'(0) = -1, \quad y''(0) = -16, \quad y'''(0) = -8.$$

Step 1: Write and factor the characteristic equation. The characteristic equation is:

$$r^4 + 4r^2 = 0.$$

Factor out r^2 :

$$r^2(r^2 + 4) = 0.$$

Step 2: Build the root ledger. The factor r^2 gives $r = 0$ with multiplicity 2. It contributes:

$$1, \quad x.$$

The factor $r^2 + 4$ gives $r = \pm 2i$. It contributes:

$$\cos(2x), \quad \sin(2x).$$

The ledger contains $2 + 2 = 4$ modes.

Step 3: Write the complete general solution. Attach one constant to each mode:

$$y = c_1 + c_2 x + c_3 \cos(2x) + c_4 \sin(2x).$$

Step 4: Differentiate three times. Differentiate once:

$$y' = c_2 - 2c_3 \sin(2x) + 2c_4 \cos(2x).$$

Differentiate again:

$$y'' = -4c_3 \cos(2x) - 4c_4 \sin(2x).$$

Differentiate a third time:

$$y''' = 8c_3 \sin(2x) - 8c_4 \cos(2x).$$

Each initial condition now has a matching formula.

Step 5: Apply $y(0) = 6$. Substitute $x = 0$ into the function formula:

$$6 = c_1 + c_2(0) + c_3 \cos 0 + c_4 \sin 0.$$

In $6 = c_1 + c_2(0) + c_3 \cos 0 + c_4 \sin 0$, use $\cos 0 = 1$ and $\sin 0 = 0$:

$$c_1 + c_3 = 6.$$

Step 6: Apply $y'(0) = -1$. Substitute $x = 0$ into the first-derivative formula:

$$-1 = c_2 - 2c_3 \sin 0 + 2c_4 \cos 0.$$

Therefore:

$$c_2 + 2c_4 = -1.$$

Step 7: Apply $y''(0) = -16$. Substitute $x = 0$ into the second-derivative formula:

$$-16 = -4c_3 \cos 0 - 4c_4 \sin 0.$$

Therefore:

$$-4c_3 = -16.$$

Divide by -4 :

$$c_3 = 4.$$

Step 8: Apply $y'''(0) = -8$. Substitute $x = 0$ into the third-derivative formula:

$$-8 = 8c_3 \sin 0 - 8c_4 \cos 0.$$

Using $\sin 0 = 0$ and $\cos 0 = 1$ in this equation gives:

$$-8c_4 = -8.$$

Divide by -8 :

$$c_4 = 1.$$

Step 9: Recover c_1 and c_2 . Use $c_1 + c_3 = 6$ with $c_3 = 4$:

$$c_1 + 4 = 6.$$

Subtract 4:

$$c_1 = 2.$$

Use $c_2 + 2c_4 = -1$ with $c_4 = 1$:

$$c_2 + 2 = -1.$$

Subtract 2:

$$c_2 = -3.$$

Step 10: Write the selected solution. Insert all four constants:

$$y(x) = 2 - 3x + 4 \cos(2x) + \sin(2x).$$

Verifying The Fourth-Order Solution

For:

$$y = 2 - 3x + 4 \cos(2x) + \sin(2x),$$

the derivatives are:

$$y' = -3 - 8 \sin(2x) + 2 \cos(2x),$$

$$y'' = -16 \cos(2x) - 4 \sin(2x),$$

$$y''' = 32 \sin(2x) - 8 \cos(2x),$$

$$y^{(4)} = 64 \cos(2x) + 16 \sin(2x).$$

Substitute y'' and $y^{(4)}$ into the ODE:

$$\begin{aligned} y^{(4)} + 4y'' &= 64 \cos(2x) + 16 \sin(2x) \\ &\quad + 4(-16 \cos(2x) - 4 \sin(2x)) \\ &= 0. \end{aligned}$$

Check the initial values:

$$y(0) = 2 + 4 = 6,$$

$$y'(0) = -3 + 2 = -1,$$

$$y''(0) = -16,$$

$$y'''(0) = -8.$$

The function satisfies the equation and all four initial conditions.

Block 6 Summary

An n th-order family has n constants. Differentiate the complete family up to order $n - 1$, then apply one initial condition to its matching formula.

Keep every constant equation visible until the full system has been solved.

Reconstructing Equations And Using Known Solutions

The Goal Of This Block

The goal is to work backward from roots or known solution forms to a characteristic polynomial, a differential equation, or a guaranteed partial basis.

From Roots To A Differential Equation

Construct a monic fourth-order equation with:

$$r = -2 \quad \text{of multiplicity } 2,$$

and the conjugate pair:

$$r = 1 \pm 2i.$$

Step 1: Build the real-root factor. The repeated root -2 gives:

$$(r + 2)^2.$$

Step 2: Build the quadratic for the conjugate pair. For roots $\alpha \pm i\beta$, the real quadratic is:

$$(r - \alpha)^2 + \beta^2.$$

Use $\alpha = 1$ and $\beta = 2$:

$$(r - 1)^2 + 4.$$

Expand:

$$\begin{aligned}(r - 1)^2 + 4 &= r^2 - 2r + 1 + 4 \\ &= r^2 - 2r + 5.\end{aligned}$$

Step 3: Multiply the factors. The characteristic polynomial is:

$$(r + 2)^2(r^2 - 2r + 5).$$

First expand the repeated factor:

$$(r + 2)^2 = r^2 + 4r + 4.$$

Now multiply:

$$\begin{aligned}(r^2 + 4r + 4)(r^2 - 2r + 5) &= (r^4 - 2r^3 + 5r^2) \\ &\quad + (4r^3 - 8r^2 + 20r) \\ &\quad + (4r^2 - 8r + 20).\end{aligned}$$

Combine equal powers:

$$\begin{aligned}(r^2 + 4r + 4)(r^2 - 2r + 5) &= r^4 + (-2 + 4)r^3 \\ &\quad + (5 - 8 + 4)r^2 \\ &\quad + (20 - 8)r + 20 \\ &= r^4 + 2r^3 + r^2 + 12r + 20.\end{aligned}$$

Step 4: Translate powers into derivatives. The associated differential equation is:

$$y^{(4)} + 2y''' + y'' + 12y' + 20y = 0.$$

Step 5: Write the general solution. The repeated real root contributes e^{-2x} and xe^{-2x} . The conjugate pair contributes $e^x \cos(2x)$ and $e^x \sin(2x)$. Therefore:

$$y = e^{-2x}(c_1 + c_2x) + e^x(c_3 \cos(2x) + c_4 \sin(2x)).$$

What A Weighted Exponential Reveals

Suppose a constant-coefficient homogeneous equation has the solution:

$$x^2 e^{3x}.$$

The factor x^2 shows that $r = 3$ has multiplicity at least 3.

Therefore the equation must also admit:

$$e^{3x} \quad \text{and} \quad xe^{3x}.$$

The guaranteed chain is:

$$e^{3x}, \quad xe^{3x}, \quad x^2 e^{3x}.$$

If the equation is exactly third-order, this chain is complete. If the equation has higher order, additional roots remain unknown.

What A Weighted Trigonometric Solution Reveals

Suppose a real-coefficient equation has the solution:

$$xe^{-x} \cos(4x).$$

The exponential and frequency reveal:

$$r = -1 \pm 4i.$$

The factor x shows that the pair has multiplicity at least 2.

Therefore the equation must admit:

$$\begin{aligned} e^{-x} \cos(4x), \\ e^{-x} \sin(4x), \\ xe^{-x} \cos(4x), \\ xe^{-x} \sin(4x). \end{aligned}$$

These four modes form a complete basis only if the equation is fourth-order.

Recognising Insufficient Information

Suppose an eighth-order equation is known to have:

$$xe^{-x} \cos(4x)$$

as a solution.

This guarantees the four-mode chain from the previous section. An eighth-order equation, however, needs eight independent modes.

For an eighth-order equation with four guaranteed modes, the number still missing is:

$$8 - 4 = 4.$$

Without more information, the remaining roots cannot be determined uniquely. State the partial basis and identify the missing dimension rather than inventing additional modes.

Block 7 Summary

Roots determine polynomial factors, and polynomial factors determine the differential equation. Weighted modes reveal minimum root multiplicities.

Compare the guaranteed basis count with the stated order before claiming that a general solution is complete.

A Reliable Higher-Order Workflow And Common Mistakes

The Complete Workflow

Step 1: Classify the equation. Confirm that it is linear, homogeneous, and constant-coefficient. Record its order n .

Step 2: Write the characteristic polynomial. Map $y^{(k)}$ to r^k , preserving coefficients and zero gaps.

Step 3: Factor or approximate the polynomial. Use common factors, grouping, substitutions such as $z = r^2$, known roots, or numerical root methods when needed.

Step 4: Build a root ledger. Record every real root, conjugate pair, and multiplicity.

Step 5: Translate each ledger row. Use the appropriate real basis rule, including every required power of x .

Step 6: Count the modes. The total must equal n .

Step 7: Apply conditions. Differentiate the complete family as many times as needed, then apply one condition at a time.

Step 8: Verify. Check the ODE, all conditions, and the final basis count.

Common Mistake: Confusing Order And Degree

For:

$$3y^{(6)} - 2y'' + y = 0,$$

the characteristic equation is:

$$3r^6 - 2r^2 + 1 = 0.$$

It is sixth-degree, not quadratic merely because only three terms are visible.

Common Mistake: Ignoring Multiplicity

The factor $(r - 4)^3$ contributes:

$$e^{4x}, \quad xe^{4x}, \quad x^2e^{4x}.$$

Writing fewer than three modes loses dimensions from the solution space.

Common Mistake: Counting A Complex Pair As One Mode

The pair $2 \pm 3i$ contributes:

$$e^{2x} \cos(3x) \quad \text{and} \quad e^{2x} \sin(3x).$$

If the pair has multiplicity 2, it contributes four functions because both trigonometric modes also receive a factor of x .

Common Mistake: Stopping Before The Count Reaches n

Finding several valid solutions does not prove that the family is complete.

For a sixth-order equation, a proposed family with only five independent functions is missing one direction, even if every displayed function solves the ODE.

Common Mistake: Expanding Too Early

The factored polynomial:

$$(r + 1)^3 ((r - 2)^2 + 9)$$

already displays all roots and multiplicities. Expanding it before building the ledger creates extra algebra and hides the structure.

Expand only when the differential equation itself is requested.

Block 8 Summary

The dependable chain is:

ODE order $\rightarrow$ characteristic polynomial $\rightarrow$ root ledger $\rightarrow$ real basis $\rightarrow$ mode count

The mode count is the central safeguard against incomplete higher-order solutions.

Original Practice Set

Practice Problems

Problem 1: Translate Derivatives Into Powers. Write the characteristic equation for:

$$3y^{(6)} - 2y^{(4)} + 7y' - 5y = 0.$$

Include the zero-coefficient terms before giving the shortened polynomial.

Problem 2: Solve A Cubic With Distinct Roots. Find the general solution of:

$$y''' + 2y'' - 5y' - 6y = 0.$$

Problem 3: Solve A Biquadratic Equation. Find the general real solution of:

$$y^{(4)} + 5y'' + 4y = 0.$$

Problem 4: Translate Repeated Real Roots. A fourth-order equation has:

$$p(r) = (r + 1)^3(r - 4).$$

Write its general solution and verify the mode count.

Problem 5: Combine Repeated Real And Complex Roots. A fourth-order equation has:

$$p(r) = (r + 2)^2(r^2 + 6r + 13).$$

Write its general real solution.

Problem 6: Handle A Repeated Complex Pair. A fourth-order equation has:

$$p(r) = ((r - 1)^2 + 9)^2.$$

Write its general real solution.

Problem 7: Keep A Repeated Zero Root. Find the general solution of:

$$y^{(5)} + 9y''' = 0.$$

Problem 8: Solve A Third-Order Initial-Value Problem. Solve:

$$y''' - 4y' = 0,$$

subject to:

$$y(0) = 3, \quad y'(0) = 2, \quad y''(0) = 8.$$

Problem 9: Construct An Equation From Roots. Construct the monic fourth-order differential equation with:

$$r = 0 \quad \text{of multiplicity } 2,$$

and:

$$r = -1 \pm 2i.$$

Also write its general solution.

Problem 10: Infer A Complete Basis From One Weighted Exponential. A fourth-order real constant-coefficient homogeneous equation has $x^3 e^{-5x}$ as a solution. Determine its general solution and characteristic polynomial.

Problem 11: Infer A Basis From One Weighted Trigonometric Mode. A sixth-order real constant-coefficient homogeneous equation has:

$$x^2 e^x \sin(3x)$$

as a solution. Determine its general real solution.

Problem 12: Recognise Missing Information. A seventh-order real constant-coefficient homogeneous equation has $x^2 e^x \sin(3x)$ as a solution. State everything that is guaranteed and explain why the full general solution cannot be determined.

Problem 13: Build A Root Ledger. For:

$$p(r) = r^2(r-5)^2((r+2)^2+1),$$

build a root ledger, write the general real solution, and verify the order.

Worked Answers: Problems 1 To 7

Answer 1. Start from:

$$3y^{(6)} - 2y^{(4)} + 7y' - 5y = 0.$$

Display every derivative order:

$$3y^{(6)} + 0y^{(5)} - 2y^{(4)} + 0y''' + 0y'' + 7y' - 5y = 0.$$

Replace $y^{(k)}$ by r^k :

$$3r^6 + 0r^5 - 2r^4 + 0r^3 + 0r^2 + 7r - 5 = 0.$$

Removing the zero-coefficient terms from the Answer 1 polynomial gives:

$$\boxed{3r^6 - 2r^4 + 7r - 5 = 0}.$$

Answer 2. The characteristic equation for $y''' + 2y'' - 5y' - 6y = 0$ is:

$$r^3 + 2r^2 - 5r - 6 = 0.$$

Factor by grouping:

$$\begin{aligned} r^3 + 2r^2 - 5r - 6 &= (r + 1)(r^2 + r - 6) \\ &= (r + 1)(r - 2)(r + 3). \end{aligned}$$

The roots are:

$$r = -1, \quad r = 2, \quad r = -3.$$

They are distinct and real, so:

$$\boxed{y = c_1 e^{-x} + c_2 e^{2x} + c_3 e^{-3x}}.$$

Answer 3. The characteristic equation is:

$$r^4 + 5r^2 + 4 = 0.$$

Let $z = r^2$:

$$z^2 + 5z + 4 = 0.$$

The equation $z^2 + 5z + 4 = 0$ factors as:

$$(z + 1)(z + 4) = 0.$$

Thus:

$$r^2 = -1 \quad \text{or} \quad r^2 = -4.$$

The roots are:

$$r = \pm i, \quad r = \pm 2i.$$

Each simple conjugate pair contributes sine and cosine modes:

$$y = c_1 \cos x + c_2 \sin x + c_3 \cos(2x) + c_4 \sin(2x).$$

Answer 4. The polynomial is:

$$p(r) = (r + 1)^3(r - 4).$$

The root $r = -1$ has multiplicity 3, so it contributes:

$$e^{-x}, \quad xe^{-x}, \quad x^2e^{-x}.$$

The simple root $r = 4$ contributes e^{4x} . Therefore:

$$y = e^{-x}(c_1 + c_2x + c_3x^2) + c_4e^{4x}.$$

The count is $3 + 1 = 4$, matching the fourth-order equation.

Answer 5. The polynomial is:

$$p(r) = (r + 2)^2(r^2 + 6r + 13).$$

The repeated real root is:

$$r = -2 \quad \text{with multiplicity 2.}$$

It contributes e^{-2x} and xe^{-2x} .

Complete the square in the quadratic:

$$r^2 + 6r + 13 = (r + 3)^2 + 4.$$

Thus the complex roots are:

$$r = -3 \pm 2i.$$

They contribute $e^{-3x} \cos(2x)$ and $e^{-3x} \sin(2x)$. Therefore:

$$y = e^{-2x}(c_1 + c_2x) + e^{-3x}(c_3 \cos(2x) + c_4 \sin(2x)).$$

Answer 6. The polynomial is:

$$p(r) = ((r - 1)^2 + 9)^2.$$

The roots are:

$$r = 1 \pm 3i.$$

For roots $r = 1 \pm 3i$, the squared quadratic means the conjugate pair has multiplicity 2. Therefore both cosine and sine modes receive powers 1 and x :

$$y = e^x [(c_1 + c_2 x) \cos(3x) + (c_3 + c_4 x) \sin(3x)].$$

The pair contributes $2 \times 2 = 4$ real modes.

Answer 7. The characteristic equation for $y^{(5)} + 9y''' = 0$ is:

$$r^5 + 9r^3 = 0.$$

Factor out r^3 :

$$r^3(r^2 + 9) = 0.$$

The root $r = 0$ has multiplicity 3, contributing:

$$1, x, x^2.$$

The roots $r = \pm 3i$ contribute:

$$\cos(3x), \sin(3x).$$

Therefore:

$$y = c_1 + c_2 x + c_3 x^2 + c_4 \cos(3x) + c_5 \sin(3x).$$

Worked Answers: Problems 8 To 13

Answer 8. The characteristic equation for $y''' - 4y' = 0$ is:

$$r^3 - 4r = 0.$$

Factor:

$$r(r - 2)(r + 2) = 0.$$

The roots are $0, 2, -2$, so:

$$y = c_1 + c_2 e^{2x} + c_3 e^{-2x}.$$

Differentiate twice:

$$\begin{aligned}y' &= 2c_2 e^{2x} - 2c_3 e^{-2x}, \\y'' &= 4c_2 e^{2x} + 4c_3 e^{-2x}.\end{aligned}$$

Apply $y(0) = 3$:

$$c_1 + c_2 + c_3 = 3.$$

Apply $y'(0) = 2$:

$$2c_2 - 2c_3 = 2.$$

Divide by 2:

$$c_2 - c_3 = 1.$$

Apply $y''(0) = 8$:

$$4c_2 + 4c_3 = 8.$$

Divide by 4:

$$c_2 + c_3 = 2.$$

Add $c_2 - c_3 = 1$ and $c_2 + c_3 = 2$:

$$2c_2 = 3.$$

Therefore:

$$c_2 = \frac{3}{2}.$$

Substitute $c_2 = 3/2$ into $c_2 + c_3 = 2$:

$$\frac{3}{2} + c_3 = 2.$$

Subtract $3/2$:

$$c_3 = \frac{1}{2}.$$

Use $c_1 + c_2 + c_3 = 3$:

$$c_1 + \frac{3}{2} + \frac{1}{2} = 3.$$

Subtracting $3/2 + 1/2 = 2$ from both sides gives:

$$c_1 = 1.$$

The selected solution is:

$$y(x) = 1 + \frac{3}{2}e^{2x} + \frac{1}{2}e^{-2x}.$$

Answer 9. The roots are $r = 0$ of multiplicity 2 and $r = -1 \pm 2i$.

The repeated zero root gives the factor:

$$r^2.$$

The conjugate pair gives:

$$(r + 1)^2 + 4 = r^2 + 2r + 5.$$

Multiply the factors:

$$r^2(r^2 + 2r + 5) = r^4 + 2r^3 + 5r^2.$$

Therefore the monic differential equation is:

$$y^{(4)} + 2y''' + 5y'' = 0.$$

The repeated zero root contributes 1 and x . The conjugate pair contributes $e^{-x} \cos(2x)$ and $e^{-x} \sin(2x)$. Thus:

$$y = c_1 + c_2x + e^{-x}(c_3 \cos(2x) + c_4 \sin(2x)).$$

Answer 10. The known solution is:

$$x^3 e^{-5x}.$$

The exponential gives root $r = -5$. The factor x^3 shows that this root has multiplicity at least 4.

Because the equation is exactly fourth-order, the multiplicity is exactly 4. The characteristic polynomial is:

$$\boxed{p(r) = (r + 5)^4}.$$

The four-mode chain is:

$$e^{-5x}, \quad xe^{-5x}, \quad x^2 e^{-5x}, \quad x^3 e^{-5x}.$$

Therefore:

$$\boxed{y = e^{-5x}(c_1 + c_2 x + c_3 x^2 + c_4 x^3)}.$$

Answer 11. The known mode is:

$$x^2 e^x \sin(3x).$$

The exponential and frequency give the conjugate roots:

$$r = 1 \pm 3i.$$

The factor x^2 shows that the pair has multiplicity at least 3. A sixth-order equation has room for exactly three copies of the pair because:

$$2 \times 3 = 6.$$

Use powers $1, x, x^2$ on both trigonometric modes:

$$y = e^x [(c_1 + c_2x + c_3x^2) \cos(3x) + (c_4 + c_5x + c_6x^2) \sin(3x)].$$

Answer 12. The mode $x^2e^x \sin(3x)$ guarantees the conjugate roots $1 \pm 3i$ with multiplicity 3. Therefore six real modes are guaranteed:

$$\begin{aligned} e^x \cos(3x), \quad xe^x \cos(3x), \quad x^2e^x \cos(3x), \\ e^x \sin(3x), \quad xe^x \sin(3x), \quad x^2e^x \sin(3x). \end{aligned}$$

The equation is seventh-order, so one mode is still missing:

$$7 - 6 = 1.$$

With real coefficients, the remaining unpaired root must be real. Call it $r = \lambda$. The most that can be stated is:

$$y = e^x [(c_1 + c_2x + c_3x^2) \cos(3x) + (c_4 + c_5x + c_6x^2) \sin(3x)] + c_7e^{\lambda x},$$

where the value of λ is unknown. The full general solution cannot be determined without additional information.

Answer 13. The polynomial is:

$$p(r) = r^2(r - 5)^2((r + 2)^2 + 1).$$

The root ledger is:

Root or pair	Multiplicity	Real modes
0	2	1, x
5	2	e^{5x}, xe^{5x}
$-2 \pm i$	1	$e^{-2x} \cos x, e^{-2x} \sin x$

Attach separate constants:

$$y = c_1 + c_2x + e^{5x}(c_3 + c_4x) + e^{-2x}(c_5 \cos x + c_6 \sin x).$$

The degree is:

$$2 + 2 + 2 = 6.$$

The six basis functions match a sixth-order differential equation.

Chapter 10 Final Summary

For:

$$a_n y^{(n)} + \cdots + a_1 y' + a_0 y = 0,$$

the characteristic equation is:

$$a_n r^n + \cdots + a_1 r + a_0 = 0.$$

A real root ρ of multiplicity s contributes:

$$e^{\rho x}, xe^{\rho x}, \dots, x^{s-1}e^{\rho x}.$$

A conjugate pair $\alpha \pm i\beta$ of multiplicity s contributes both the cosine and sine chains with powers x^0 through x^{s-1} .

The chapter's final lesson is:

list every root, preserve every multiplicity, pair every complex root with its conjugate, and count the resulting real modes against the equation's order before calling the solution complete.